

Lines and Planes in $\mathbb{R}^3$

Robert O'Connell

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Lines in $\mathbb{R}^3$

A line in $\mathbb{R}^3$ is completely determined by a point and a direction vector

Vector Equation

Let $\overrightarrow{0P_0}$ be the vector joining the origin in $\mathbb{R}^3$ to the point P_0

The vector equation of the line through P_0 with direction vector $\vec{v}$ is given by

$$\vec{r}(t) = \overrightarrow{0P_0} + t\vec{v}, \quad t \in \mathbb{R}$$

All vectors that satisfy the equation have their tips lying on the line.

As t ranges from $-\infty$ to ∞ the tips of the vectors trace out the line

With Coordinates

Let $P_0 = (x_0, y_0, z_0)$, $\vec{v} = \langle a, b, c \rangle$

$$\begin{aligned}\vec{r}(t) &= \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle \\ &= \langle x_0 + ta, y_0 + tb, z_0 + tc \rangle\end{aligned}$$

A vector $\langle x, y, z \rangle \in \mathbb{R}^3$ lies on the line if and only if

$$x = x_0 + ta$$

$$y = y_0 + tb$$

$$z = z_0 + tc$$

Planes in $\mathbb{R}^3$

A plane in $\mathbb{R}^3$ is completely determined by a point and a normal vector (perpendicular to the plane)

Equation

We wish to describe the plane in $\mathbb{R}^3$ that passes through the point $P_0(x_0, y_0, z_0)$ and that has $\vec{n}\langle a, b, c \rangle$ as the normal vector

Let $P(x, y, z)$ be any other point in the plane

Let $\overrightarrow{PP_0}$ be the vector that joins P to P_0

$\overrightarrow{PP_0}$ lies in the plane

Thus $\overrightarrow{PP_0} \cdot \vec{n} = 0$

$$\langle x - x_0, y - y_0, z - z_0 \rangle \cdot \langle a, b, c \rangle = 0$$

$$ax + by + cz = ax_0 + by_0 + cz_0$$

Note

In this form the coefficients of x, y, z form the components of the normal vector